

Computer Networks: Quick Study Notes

A compact, original study note for demonstrating document upload, RAG question answering, summarization, and AI quiz generation in UniPrep.

Main topics	OSI and TCP/IP, network devices, IP addressing, common services, security
Suggested use	Upload to UniPrep, ask questions, generate a summary, then generate a quiz
Difficulty	Beginner to intermediate

Learning goal: Understand how devices communicate across networks, how protocols divide responsibilities, and how common Internet services operate.

1. What Is a Computer Network?

A computer network is a group of connected devices that exchange data and share resources. Devices may include computers, phones, servers, routers, printers, and sensors. Networks can be small, such as a home Wi-Fi network, or very large, such as the Internet.

- **LAN (Local Area Network):** Covers a limited area such as a room, office, or campus building.
- **WAN (Wide Area Network):** Connects networks over large geographic areas. The Internet is the largest WAN.
- **Client:** A device or application that requests a service.
- **Server:** A device or application that provides a service, such as a website, file, or database.

2. Network Devices

Device	Main function
Switch	Connects devices inside a LAN and forwards frames using MAC addresses.
Router	Connects different networks and forwards packets using IP addresses.
Wireless access point	Allows wireless devices to join a wired network.
Modem	Converts signals so a local network can communicate with an Internet service provider.
Firewall	Applies security rules to allow or block network traffic.

3. OSI and TCP/IP Models

Networking models divide communication into layers. Each layer performs a specific job and provides services to the layer above it. This separation makes networks easier to design, troubleshoot, and update.

OSI layer	Purpose	Examples
7. Application	Provides services used by applications	HTTP, DNS, SMTP
6. Presentation	Data format, compression, encryption	TLS formatting concepts
5. Session	Starts, manages, and ends sessions	Session control
4. Transport	End-to-end delivery and reliability	TCP, UDP
3. Network	Logical addressing and routing	IP, routers
2. Data Link	Local delivery using frames and MAC addresses	Ethernet, switches
1. Physical	Transmits bits through media	Copper, fiber, radio

TCP/IP model: In practical Internet networking, the layers are commonly grouped into Application, Transport, Internet, and Network Access layers.

4. TCP and UDP

TCP and UDP are transport-layer protocols. They use port numbers to deliver data to the correct application on a device.

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Acknowledgments, retransmission, and sequencing	No built-in delivery guarantee
Overhead	Higher	Lower
Typical uses	Web pages, email, and file transfer	Streaming, voice, DNS queries, and online games

5. IP Addressing

An IP address identifies a network interface. IPv4 addresses contain 32 bits and are usually written as four decimal numbers, such as 192.168.1.10. A subnet mask separates the network portion from the host portion.

- **Private IPv4 ranges:** 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.
- **Default gateway:** The router address used to reach other networks.
- **Subnetting:** Divides one network into smaller logical networks.
- **IPv6:** Uses 128-bit addresses and provides a much larger address space than IPv4.

Example: /24 Network

For 192.168.10.55/24, the first 24 bits represent the network. The network address is 192.168.10.0, the broadcast address is 192.168.10.255, and normal host addresses range from 192.168.10.1 to 192.168.10.254.

6. Common Network Services

Service	Purpose	Common port
DNS	Translates domain names into IP addresses	53
DHCP	Automatically assigns IP configuration to clients	67/68
HTTP	Transfers web content	80
HTTPS	Transfers encrypted web content	443
FTP	Transfers files	21
SMTP	Sends email between systems	25
SSH	Secure remote command-line access	22

7. How DNS and DHCP Work

DNS process

- A user enters a domain name, such as example.com.
- The client asks a DNS resolver for the corresponding IP address.
- The resolver may use cached data or query other DNS servers.
- The IP address is returned to the client, which can then contact the destination server.

DHCP process

- The client broadcasts a discovery message.
- A DHCP server offers an available IP configuration.
- The client requests the offered address.
- The server confirms the lease and provides details such as subnet mask, gateway, and DNS server.

8. Basic Network Security

Network security protects confidentiality, integrity, and availability. A secure design uses multiple controls rather than relying on one device or password.

- **Authentication:** Confirms the identity of a user or device.
- **Authorization:** Determines what an authenticated user is allowed to do.
- **Encryption:** Protects data from unauthorized reading.
- **Firewall rules:** Permit required traffic and block unnecessary traffic.
- **Updates and patching:** Fix known vulnerabilities.
- **Network segmentation:** Limits how far an attacker can move inside a network.

9. Troubleshooting Method

A structured troubleshooting process reduces guesswork. Start with simple physical checks, then move upward through addressing, routing, and application services.

Step	Example check
1. Physical	Power, cables, Wi-Fi signal, link lights
2. Local configuration	IP address, subnet mask, default gateway
3. Reachability	Use ping to test local and remote hosts
4. Routing	Use traceroute or tracert to inspect the path
5. Name resolution	Use nslookup or dig to test DNS
6. Application	Check ports, service status, firewall, and logs

10. Key Takeaways

- Switches primarily forward frames inside a LAN; routers forward packets between networks.
- TCP emphasizes reliable delivery, while UDP emphasizes speed and low overhead.

- DNS resolves names, DHCP assigns network settings, and HTTP/HTTPS deliver web content.
- IP addresses identify interfaces, while subnet masks identify the network boundary.
- Layered models help explain communication and isolate faults during troubleshooting.

Suggested UniPrep demo prompts:

1. Summarize this document in five points.
2. What is the difference between a switch and a router?
3. Explain how DHCP works.
4. Create five multiple-choice questions from this note.
5. Which protocol should be used for reliable file transfer, and why?